

MIR SHAHAAN ABBAS ABEDI

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EDUCATION

Bachelor of Engineering (B.E.) – Artificial Intelligence and Data Science

MS Ramaiah Institute of Technology, Bengaluru | CGPA: 9.66 / 10 | Oct 2023 – Present

Pre-University (12th Grade) - Physics, Chemistry, Maths

Gem International Composite PU College, Gauribidanur | 95.17% | May 2021 – Mar 2023

COURSE WORK

Data Structures & Algorithms (DSA) | Database Management Systems (DBMS) | OOP Concepts

Web Development | Data Science | Big Data Analytics | Computer Networks (DCN) | Cloud Computing

TECHNICAL SKILLS

Languages: Java, C, Python, JavaScript

Frameworks & Libraries: React, Node.js, Express.js

Web Technologies: HTML5, CSS3

Databases: MySQL, MongoDB

Developer Tools: VS Code, Jupyter Notebook, Google Colab, Git, GitHub

Data Tools: Power BI, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

PROJECTS:

RapidAid – Vision-Based Accident Detection & Victim Analysis | April 2026

Technologies: Python, YOLOv8, OpenCV, Google Colab, NumPy, FastAPI

- Engineered a real-time CCTV accident detection pipeline combining YOLOv8 with custom IoU collision math to instantly isolate crashed vehicles without requiring custom-labeled datasets.
- Developed a victim classification system utilizing pose detection and spatial filtering to accurately differentiate fallen victims from active bystanders in chaotic crash scenes.
- Architected a multi-stage inference pipeline that processes raw frames and outputs structured JSON payloads to trigger automated emergency dispatch and PDF report generation.

GPU-Accelerated Parallel FFT Audio Filter | November 2025

Technologies: Python (FastAPI, CuPy, NumPy), CUDA, JavaScript, HTML5 Canvas

- Architected a high-performance audio processing engine using CuPy/CUDA to perform parallel Real FFT (rFFT) operations, enabling real-time spectral filtering with automatic CPU fallback.
- Developed an interactive spectral editor using HTML5 Canvas that allows users to "paint" frequency masks to selectively attenuate audio components with instant visual and auditory feedback.
- Engineered a stateful FastAPI backend to manage batched audio frames and overlap-add synthesis, facilitating seamless reconstruction of filtered signals from user-defined frequency masks.

Subscription Burnout Segmentation | December 2025

Technologies: Python, Scikit-learn, Pandas, PySwarms, Matplotlib, Joblib

- Developed a machine learning pipeline using K-Means clustering optimized with Particle Swarm Optimization (PSO) to segment 5,600+ unique videos into 4 distinct engagement burnout patterns.
- Engineered time-series features (Slope, Peak Drop, Initial Engagement) from a dataset of 40,000+ records to identify high-risk content and quantify momentum loss.
- Architected an automated batch inference pipeline to generate daily sustainability reports and diagnostic visualizations for data-driven content strategy.

Chandrayaan 3 Data Analysis(Moon Mission) | May 2025

Technologies: React, Material UI, CSS, OpenWeather API, JavaScript

- Analyzed Chandrayaan-3 mission data using Python to explore mass, power, and trajectory.
- Built visualizations and an interactive dashboard to surface key insights and support storytelling in Power BI
- Cleaned and transformed numeric data with regex and Pandas to improve analysis quality